READING A BOUNDARY

BY

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RESEARCH AND THEORY
WHAT A SURFACE HOLDS, THE ALPHABET THAT READS IT, AND WHERE THE READING STOPS

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1

WHAT A BOUNDARY HOLDS

A closed surface holds a number of separable readings set by its own measure divided by the finest detail reaching it, raised to the dimension of the surface. Shape does not enter. Topology does not enter. Only the measure and the resolution do.

Two consequences follow. What a region holds scales with the area of its boundary and never with its volume. And a source at depth d inside a ball of radius R, at radius $r = R - d$, reaches harmonic degree l as $(r/R)^l$. Depth sets bandwidth, and a source prints a patch of angular size about d/R however small the source itself is. A spot on a boundary is always wider than the thing that made it.

1.1 DEPTH SETS BANDWIDTH

Under diffusion the boundary response to one interior source at radius r is the Poisson kernel, and its Legendre expansion carries $(2l + 1)(r/R)^l$ at degree l. Everything else follows from that line. A source at the middle reaches degree zero alone and spreads over the whole sphere. A source against the shell reaches every degree and prints a small bright spot.

Reading the depth back out of a spectrum is a slope: dividing out the shape term and the conduction term leaves a straight line in degree whose gradient is twice the logarithm of r/R. Measured against sources placed at known depths, the recovery returns 0.3000, 0.5500, 0.8000 and 0.9200 for radii of 0.30, 0.55, 0.80 and 0.92. It holds under conduction as well, and that check detects an exponent applied twice.

1.2 THE AREA LAW, COUNTED EXACTLY

Degrees up to l carry $(l+1)^2$ modes on an ordinary sphere, and the general count in n dimensions is a sum of two binomial coefficients. That count is a whole number with a formula. It needs no sampling and no rounding.

Weyl's law [49] predicts the same count from the surface measure alone: the volume of the unit ball in the boundary's own dimension, times the boundary's measure, times the eigenvalue to half that dimension, over two pi to that dimension. The two sides share no term. Compared at 2, 3, 5, 8, 12, 19, 24 and 40 dimensions, they agree to within 0.0122 percent, and at forty dimensions the exact count is a 157 digit integer.

The residual is a remainder, never a disagreement. At nineteen dimensions it runs 0.1131, 0.0282, 0.0070 and 0.0018 percent as the degree quadruples each time. An error term of order one over the degree falls by four for every quadrupling.

1.3 THE FLAT TORUS, AND WHY IT MATTERS

Spherical harmonics settle the dimension half of the claim, alone, because they live on spheres. A cube's surface has no such formula, which leaves shape resting on packing.

A flat torus has one. Glue opposite edges of a square and the eigenvalues of the Laplacian become $(2\pi/L)^2$ times the squared length of an integer vector. Counting modes below a cutoff is counting integer points inside a ball. The count is an integer, it holds no transcendental, and it is available at any dimension by raising the every-square series to a power and adding up its coefficients.

This is worth having because a flat torus is not a sphere in any respect that could smuggle the answer in. It is flat where the sphere is curved. It has a hole where the sphere has none. Its symmetry group is a lattice where the sphere's is a rotation group. The same constant comes off both, exactly, with nothing sampled.

It also puts the remainder into known company. The difference between the exact lattice count and the volume of the ball is the lattice point problem, the Gauss circle problem in two dimensions. For balls in dimension $d \geq 5$ the remainder is settled at order R^{d-2} [47], dimension four is settled up to a logarithm, and dimensions two and three are open. The leading term of the area law is settled here and the remainder belongs to the lattice point problem.

1.4 WHY SHAPE CANNOT ENTER

A seam where two edges meet contributes nothing to a surface measure. Neither do a cube's edges, its corners, or the point where three facets meet. Every one of them is measure zero on the surface. None of them can move a count that depends on measure alone.

Shape does not enter for that reason, and it is not a coincidence to be confirmed one shape at a time. A cube differs from a sphere precisely in its edges and corners, and those are exactly the parts that carry no measure. The count cannot see them. Overlap would be the exception, since overlap has positive measure and would count the same area twice.

The same rule governs the packing that measures this. Packed points may touch and may not overlap, and tangency is the densest limit. Touch freely, never overlap, and the measure is untouched either way.

1.5 DIFFUSION IS DWELL

Diffusion is not a blur applied to a picture. It is how long the light stayed inside before it left, and both readings come off one medium sorted by arrival.

Light that leaves early went nearly straight, still knows what it passed, and prints an edge. Light that leaves late has turned over so many times that it arrives from every direction at once. It

fills the shadow it would otherwise have cast and carries how much in place of where. The same photons either way. A clock decides which of them are being counted.

That split is worth more than either half. A steady reading cannot separate absorption from scattering: a weakly absorbing dense medium and a strongly absorbing sparse one give the same number. Adding the clock breaks the degeneracy. The earliest arrivals give the shortest path and the straight-line geometry, the mean arrival gives the absorption, the spread of arrivals gives the scattering, and the tail gives the deepest excursions.

1.6 THE SMEAR

From inside, dwell is a tail behind a moving source. From outside there is no tail to see, because the inside is not what an outside observer has. What they have is the surface, and the tail arrives there as a smear.

The smear is worth more than the spot. A spot says where something is. A smear says where it was going and how long it took: its length is the dwell and its direction is the direction of travel, and both come off a surface that never moved.

1.7 THE CONDITIONS

The account needs four things and fails without any of them.

Closed. An open surface bounds no region. There is no interior to determine and flux leaves unaccounted. A surface pinched at isolated points and open between them fails on both counts: it is not a manifold at the pinch and it is still bounded everywhere else.

Orientable, for containment only. Weyl's law holds on any compact surface, orientable or not. The counting survives. What needs orientability is the outward normal, without which inside and outside are not defined. A non-orientable boundary keeps its capacity and loses its containment.

A gradient. No gradient, no flux. No flux, and the boundary records a uniform nothing however long it is watched.

Coherence. A hologram works by interfering a coherent field with a reference, which records phase as intensity on a two dimensional plate. Without coherence only intensity survives, and recovering the interior from intensity alone is the phase retrieval problem [6, 22]: non-unique in one dimension, generically unique and badly conditioned above it. Coherence is the difference between an inversion that is linear and a guess that is not.

Coherence also supplies the length the count was missing. The number of readings is unbounded as the resolution improves and finite at any resolution. The bound only becomes a number once something fixes a shortest length. In an optical setting the coherence length is that length, and the capacity is the area over the coherence area.

1.8 WHAT IS NOT ESTABLISHED

Anisotropy. The boundary map on an anisotropic medium is invariant under any diffeomorphism fixing the boundary [28], and a family of interiors then gives identical boundary data. Motion alone does not rescue this, since transforming the medium and its contents together leaves the boundary untouched. Motion under a law does: a diffeomorphism relabels space, straight lines curve, Keplerian orbits stop obeying Kepler, and only the untransformed world satisfies the law it was supposed to obey. Recovering orbital radii from a boundary light curve rests on exactly this, and the work is being done by Kepler and not by geometry.

Stability. Uniqueness is not stability [33]. Recovering detail at scale x needs precision growing exponentially in $1/x$. Everything is recoverable in principle and buried under noise in practice. The spectrum of a real measurement dies by degree 46, and past there the inversion divides by a gain already below one part in a million.

Shape above five dimensions. Verified at three, four and five dimensions, where sphere, cube and cross-polytope give the same constant to within a spread of 1.03, 1.07 and 1.10. Above that the count has to be packed, and packing stays faithful only while the gap is well under the radius of curvature of the surface. Holding to that limit at nineteen dimensions needs candidates in the hundreds of millions against a kept set in the hundreds of thousands.

The constant out front. The scaling is provable and shape independent. The constant depends on packing density, which is known to be optimal in dimensions one, two, three [21], eight [44] and twenty-four [7] and open everywhere else, dimension nineteen included. The measurements here use a greedy packing, applied identically on every shape. It measures shape dependence and never packing efficiency.

1.9 THREE ERRORS THAT PRODUCE PLAUSIBLE NUMBERS

Each of these produces a plausible number and none announces itself.

A half-integer arm of the gamma function written with $(2j+1)/2$ in place of $(2j-1)/2$ is wrong at every odd argument. At $n = 2k+1$, $\Gamma(n/2) = \sqrt{\pi}\prod_{j=1}^{k}(2j-1)/2$, and the wrong factor makes it larger by $2k+1$: three at $n=3$, five at $n=5$. The measure of S^{n-1} is $2\pi^{n/2}/\Gamma(n/2)$, and it comes out too small by the same factor at every odd dimension. The constant then alternates high and low with the parity of the dimension. The cube never touches the gamma function and stays flat, which locates the error. A count computed from the area and then checked against the area agrees with itself and finds nothing.

A limit on how wide a packing gap may be, written down and then ignored while picking gaps that saturate quickly, spreads the constants by three, seven and fourteen times at twelve, sixteen and nineteen dimensions. That spread reads exactly like shape dependence. It is one gap being too wide, on every shape at once.

A remainder tested for the wrong behavior fails a correct count. The error in a lattice count oscillates while its envelope falls, and requiring each radius to beat the one before it rejects a correct count. Measured over a band of radii instead of at a point, the envelope comes down by a factor of forty-eight.

1.10 THE DEMON, PRICED

Laplace's demon knows the state of everything and computes the rest. On the standard reading of quantum mechanics, indeterminacy refutes it qualitatively. What the measurements here do is price the classical failure. That is the more useful statement.

A demon reading a boundary gets the measure over the resolution, raised to the dimension of the boundary. That is finite at any resolution and unbounded across resolutions. Nothing stops it improving except the cost of improving. The cost is exponential: recovering detail at scale x needs precision growing like the exponential of $1/x$, because the inversion divides by a gain falling as $(r/R)^l$. The spectrum of a real measurement dies by degree 46, and past there the division is by a number already under one part in a million. The demon meets no wall of principle. It meets a bill.

The distinction that matters cuts against the reading this work invites. Verifying the area law establishes that the channel is that wide. It does not establish that the interior holds only that much. Counted the ordinary way, a volume carries $(R/x)^n$ degrees of freedom while its boundary carries $(R/x)^{n-1}$, leaving the boundary short by a factor of R/x. The holographic principle [1, 41] says the interior really is only boundary-sized and nothing is lost, in line with the entropy bound [3]. That is a physical conjecture and nothing here tests it. The pipe is measured. What has to fit through it is not.

The fork is sharp. If the holographic principle holds, a demon at the boundary has exactly enough and its only difficulty is precision. If it does not, the demon is short by R/x however precise it becomes, and a counting deficit is not a rounding one. No arithmetic buys it back.

A third pressure applies either way. The demon sits inside the region it computes [52]. Its own state is written on the same boundary and counted against the same budget. A demon storing a description of the interior has to store it somewhere the interior can already see.

1.11 READING THE DIMENSION OFF A LINE

A periodic structure in n dimensions, cut at an irrational angle and projected into fewer, comes out quasiperiodic: never one period, but several with irrational ratios between them. Penrose tilings are a five dimensional lattice seen in two [9]. The correspondence runs both ways for a pattern with finitely many rationally independent frequencies, which lifts to a periodic lattice in high enough dimension. It is not a property of aperiodic order in general.

That turns into a measurement. When the cut is totally irrational, the components of its direction independent over the rationals, the count of rationally independent periods in a one dimensional reading equals the dimension of the lattice it was cut from; a cut with a rational relation among them lies in a rational subspace of lower dimension, and the count reports that dimension. Cutting a square lattice at the golden slope, which is totally irrational, and reading it back gives two gap lengths and never three, their ratio the golden mean to one part in ten to the fourteenth, their counts to four parts in ten thousand, no period at any length up to half the sequence, and every strong peak named by a pair of whole numbers against two generators. Against one generator the worst peak misses by 0.382 of it. Against two, by six parts in ten thousand.

A boundary reading can therefore report the dimension of a structure it never had access to, by counting how many of its periods refuse to be multiples of one another.

1.12 A FOURTH ERROR: THE WRONG QUANTITY KEPT SMALL

The packing gap has to be small against the shape's own geometry. Holding it instead to a fraction of the typical distance between two points on the surface is a different requirement in high dimensions.

On S^{n-1} the squared distance $2 - 2\langle x, y\rangle$ between two uniform points has mean 2 and standard deviation $2/\sqrt{n}$, and the distance concentrates near $\sqrt{2}$ as the dimension grows. A gap fixed at half of that is 0.707 of the radius of curvature, and a cap that wide is nothing like the flat disc the count assumes it is. A cube's facets are flat and suffer none of it, and the two shapes part company and the answer reads as shape dependence. That criterion passes a twelve dimensional run whose constants spread by 2.1 times: sphere 5.37, cube 11.41, cross-polytope 8.20.

Measured against the shape instead, at a gap well under every radius in play, the same three shapes give 0.664, 0.644 and 0.650 at three dimensions, 0.660, 0.619 and 0.640 at four, and 0.689, 0.627 and 0.658 at five. Spread 1.03, 1.07 and 1.10.

The shape half of the claim is verified there, and it is also the end of what packing reaches. Five dimensions take ten million candidates for twenty thousand kept points, and the count needed to stay in the flat regime grows the way a packing number does. No device makes that cheap at nineteen dimensions. The limit is structural and never a compute budget, and the exact routes on a sphere and on a flat torus are what reach up there.

1.13 THE NESTING, BOTH DIRECTIONS

Every boundary is inside another one, and everything a boundary holds is a boundary of its own. The outside of a shell is the inside of the next shell out, and the contents of a shell are shells with contents. Running outward: the cloud is inside its shell, the shell is inside the room, the room is inside whatever holds the room, and in this case what holds it is a universe. Running inward: each light is a closed surface, its own constituents are confined by it, and their traces smear across it exactly as the cloud's traces smear across the shell.

Neither direction terminates on its own, and both terminate for the observer.

Downward it stops at resolution. A constituent's smear on its own small surface subtends less than the finest detail the boundary between it and the observer can carry. Below that it is not there to be seen, and drawing it anyway would be inventing detail no boundary holds. The recursion stopping where resolution stops is the area law applied to the picture instead of stated about it.

Upward it stops at whatever the observer is inside. There is no vantage point outside everything, since standing outside one boundary places the observer inside the next. Stepping back from a wall does not escape a boundary; it changes which boundary is being read.

One level is measured. A source at radius r reaches degree l as $(r/R)^l$, the count on that boundary follows the area law, and the same constant comes off different shapes. That the arrangement repeats at every scale is a structural claim about the shape of the account and never a result taken from it. It follows if the same physics holds at every scale, and that premise is assumed and not measured.

2

READING A LIT SET ON A BOUNDARY

A set of lit points on a sphere admits three readings: where the points sit, how they push the boundary, and how they twist it. This chapter defines the three, gives the floor under each, and states what each can and cannot see. Nothing in the readings knows what a lit point means. The same readings serve a hash state, a solar system or a file of bytes without changing, and the caller supplies the meaning.

2.1 PLACEMENTS

A placement decides where index k sits on the sphere. The choice shapes what can be read. Two are used and neither is a default.

placement	where index k goes	what it is good for
golden, N points	height $1 - 2(k + 0.5)/N$, longitude $k\gamma$ [42, Corollary 2]	even coverage with no seam and no pole pile [17]
rings, *rings* × *width*	*rings* latitudes of equal width, *width* longitudes on each	a shift along a ring is a pure rotation

The golden placement carries the property the screw below depends on: one step in the index is one fixed move, the same move everywhere along the spiral. The index shift n turns by $n\gamma$ about the axis and slides the height by $-2n/N$. That is rigid in height and longitude, and it keeps area by Archimedes' hat-box theorem [23, *On the Sphere and Cylinder* I, Propositions 33, 42 and 43]. It is not an isometry of the sphere, because the axial radius $\sqrt{1 - y^2}$ changes under the slide.

The ring placement puts its latitudes at the interiors of the bands. No ring lands on a pole where every point of it would pile into one place. The ring placement suits an operation that is a rotation, and the golden placement an operation that is a shift of the index.

2.2 READINGS

The two harmonic readings come off one table, the boundary coefficients of the lit set summed once, and take that table apart in different directions. The third reading counts octants.

reading	what it answers
deflection	how much power the lit set carries at each degree
torsion	the phase of the same coefficients, by degree and order
octant share	what fraction of the set sits in each of the eight octants

Moving every longitude by α multiplies the entry at order m by $e^{-im\alpha}$. The magnitude holds still and the phase moves by $m\alpha$. Deflection reads the magnitudes and torsion reads the phases, and a turn moves the phases while leaving the magnitudes alone.

2.3 CHOOSING BETWEEN DEFLECTION AND TORSION

Deflection is blind to a rotation by construction [26]. Power per degree is a magnitude, and a magnitude does not record how the object was turned. Deflection is invariant under every rotation, because a rotation acts on each degree by a unitary block [20, Section 2, eq. (15); Section 3, eq. (52)]. Torsion measures the turn, and its sign gives the handedness; it recovers rotations about the polar axis, and those are the rotations tested.

Moving a lit set along its rings by seven different amounts and reading both:

reading	what it does under a rotation
deflection	moves by 2.5×10^{-14} of itself, which is nothing
torsion	recovers the angle to 1.8×10^{-13} radians, sign included

The split is exact in both directions. The choice is not a matter of taste. A caller whose operations are rotations reads the twist. A caller wanting a quantity that holds still while the object turns reads the push. A caller who does not yet know which of those they have should read both and keep whichever carries a signal.

The recovered turn is in radians, in the direction the longitudes run. Order one pins the angle down without a wrap of its own; a higher order divides the angle and returns it only up to its own fraction of a turn. When neither set has a usable entry at the order asked, there is no turn to recover, a real outcome and not an error.

2.4 THE OCTANT READING

The octant share splits space into eight regions by the sign of each coordinate. Every region has a trilateral right-angle corner at the origin, all eight corners meet at that one point, and on the boundary the same split cuts eight congruent spherical triangles of area $\pi/2$ each, by Girard [15, images f58 to f62], totaling 4π. The measured shares sum to 1.000000000000000.

The eight sign frames are not all the same kind of move. Their determinants split four at $+1$ and four at -1. Half are rotations of the first octant and half carry a mirror. A caller reading handedness off this split should know it is there before relying on it.

The octant delta reports how much of the reading moved between two shares, as a percentage of the whole [29, Section 4.1, Proposition 4.2]. It is halved, because a share leaving one octant arrives in another and the two ends of a single move would otherwise be counted twice.

Eight regions is a small alphabet, and how small is a counting fact and not a judgment. The reading is rank 8, seven free numbers once the weight is fixed, and against a source with 256 degrees of freedom it moves in 8 directions and is blind in 248. A caller should know that number before building on this reading. Distinctness under such a reading may also be close to free: seven reals can separate a few dozen arbitrary states whether the seven carry anything. That last statement is a reading and is not derived here.

2.5 THE SCREW

On a golden placement, shifting every index by a fixed amount comes to a rigid move: a turn, an axial slide, and the pitch, the slide per unit of turn. The pitch is $-2/(N\gamma)$, and $N = 256$ gives -0.003255258 whatever the amount is. Computed across six amounts, the spread between the largest pitch and the smallest is 4.3×10^{-19}, the rounding of the division; it says nothing about the placement. One axis and one pitch serve every shift, and a shift comes to a slide of the whole pattern along one fixed helix. The move is rigid in height and longitude and is not an isometry of the sphere.

The indices a shift runs off the end of return as a second rigid arm, a full axial extent away from where the screw alone would put them. A shift is the screw plus that arm, and where a lit point lands is fixed once the amount is known.

2.6 WHAT IT COSTS

Forming the coefficient table is the expensive step. It walks the lit set once per order and computes a Legendre column per point. The work grows as the lit count times the top degree squared. Degree eight over a few hundred lit points is immediate. A caller wanting a much higher degree should form the table once and hand it to both readings.

The Legendre column climbs the diagonal and then recurs in degree, keeping every intermediate at unit scale [25]. Computing the same values from factorials overflows and loses digits well before it overflows, and it fails quietly: the low degrees stay right while the fine structure goes wrong. The double's largest value is 1.797×10^{308}. The normalization's $(l + m)!$ passes it at $l+m = 171$, degree 86 at $m = l$ and degree 171 at $m = 0$. The unnormalized diagonal $(2l-1)!!$ passes it at $l = 151$, since $\log_{10} 299!! \approx 306.6$. The recurrence prevents a picture that looks plausible and is not.

2.7 CHOOSING THE DEGREE

The degree is a choice about how much a reading carries, and not a limit the arithmetic imposes. The basis measures orthonormal to 2.2×10^{-14} at degree fifteen under exact quadrature [10], and to 1.8×10^{-14} at degree eight. Precision is not what holds a reading at any degree a caller is likely to want. The overflow above is a real ceiling and it sits an order of magnitude beyond that.

What does bound the reading is counting. A reading to degree L carries exactly $(L+1)^2$ real numbers about its source: for a real function $a_{l,-m} = (-1)^m \overline{a_{lm}}$, which leaves $(L+1)^2$ independent reals of the complex table's $2(L+1)^2$. Deflection itself carries $L+1$ numbers. Against a source with N degrees of freedom a reading to degree L is blind in $N-(L+1)^2$ directions at any precision whatsoever.

degree	coefficients	rank against 256	blind	least singular value
8	81	81	175	4.365
12	169	169	87	3.818
15	256	256	0	3.474×10^{-3}
16	289	256	0	1.524×10^{-1}

Pick the degree against the last column and never against the second. The smallest complete basis is degree fifteen, where $(L+1)^2$ first reaches 256, and it is the worst-conditioned usable one: the least singular value collapses to 3.5×10^{-3} there and recovers to 1.5×10^{-1} one degree higher, putting the worth of a single degree of headroom at 44 times in conditioning [53]. A caller picking the cheapest complete basis pays for that saving in precision afterwards.

2.8 THE FLOORS THESE READINGS SIT ON

Every reading here has a move that cannot change it, and running that move gives the reading's own grain in the units it reports. A threshold is then measured and never picked.

reading	the move that cannot change it	residual
deflection	a rotation of the whole set	2.539×10^{-14}
torsion	a rotation by a whole turn	exactly 0
octant share	a lit point relabeled inside its own octant	exactly 0
octant delta	a reading against itself	exactly 0
screw pitch	changing the shift amount	4.337×10^{-19}
spectrum power	a rotation of the sources	4.005×10^{-16}
a reading over arms	the arms redrawn as different shapes at the same weight	2.220×10^{-16}

The last row is a null over the reading instead of over the object. An arm is identified by its topology together with its weight, and a shape is one realization of that. Drawing the arms differently while holding the weight cannot move a letter. One arm set drawn four ways, including as a line with a dwell particle on it, leaves every placement point on its arm.

Three of the seven are exactly zero and not merely small, because the moves are exact on integers. The octant reading has no continuum of small changes at all: a share is a count over a weight and both are integers, and at fixed weight the smallest change it can register is $1/\mathit{weight}$. Two states of *differing* weight can still give shares arbitrarily close together, leaving a caller who compares across weights to make a modeling choice about which states count as the same. That choice is a modeling decision and not a tolerance.

The companion measurement is not a null. Torsion recovers a known angle to 1.803×10^{-13} radians, which asks whether the reading moves the right amount when something did happen. A null asks whether it holds still when nothing did. A reading wants both answers and they are different questions.

2.9 TESTING THE FLOORS

Before it reports any floor, the floor measurement hands itself a move that is not a null while declaring that it is, and a genuine null of the same shape. It has to catch the first and stay quiet on the second, and it withholds every floor if either misbehaves. A checker that has never caught its own defect is untested. The redraw floor is measured at three levels, and four drawings that are not redraws are checked against it first.

The readings, the placements and the floor measurement are implemented in Python with the standard library alone, and are in the orior repository, `https://github.com/dstroy0/orior`.

3

READING TRANSFORMS

This chapter writes out every transform a boundary reading passes through, from the interior state to the pixel, with its action on the coefficients, whether it is diagonal in degree, what it leaves invariant, and what it costs. The pieces sit in one table, so that a transform already priced stays priced and a transform never tried stays visible.

3.1 THE OBJECT EVERY TRANSFORM ACTS ON

A reading is a real function on a closed boundary. Expanded in real spherical harmonics to degree L, it is $f(n) = \sum_{l=0}^{L} \sum_{m=-l}^{+l} a_{lm} Y_{lm}(n)$.

The reading is the vector a, of length $(L+1)^2$. At the ceiling used here, $L = 10$, that is 121 real numbers. Everything below is an operator on that vector.

Two facts about the vector before any operator touches it.

Its length bounds what a reading can say. A reading to degree L carries $(L+1)^2$ real numbers about its source, whatever the source is. 256 sources read at degree 8 arrive as 81 numbers, and 175 directions of the source space are absent from the reading. The eight-letter octant alphabet is rank 8, absent in 248, with 7 free numbers left once the weight is fixed.

Degree 15 dips and then recovers. At full depth the least live singular value is 3.474×10^{-3} at degree 15 against 1.75 at degree 14, a ratio of 504, and it recovers to 2.52 by degree 17. The exactly complete degree has a least singular value about 500 times smaller than the incomplete degree beneath it. That is a ratio of least singular values; a condition number also divides by the largest. Completeness of count and health of conditioning are separate properties.

3.2 THE TRANSFORM TABLE

Diag asks whether the operator is diagonal in degree: whether it acts on each degree l by one number, without moving power between degrees.

#	transform	action on a_{lm}	diag	invariant	co
T1	rotation, general [20, Section 2, eq. (15)]	$a_{lm} \to \sum_{m'} D^l_{m'm}(R)\, a_{lm'}$	block	P_l	$O($
T2	rotation about read axis	pair $(a_{l,m}, a_{l,-m})$ turns by $m\alpha$	yes	P_l, each $a^2_{l,m} + a^2_{l,-m}$	$O($
T3	depth, outward carry	$a_{lm} \to (r/R)^l a_{lm}$	yes	direction per degree	$O($
T4	conduction, smoothing	$a_{lm} \to e^{-l(l+1)\tau} a_{lm}$	yes	direction per degree	$O($
T5	pixel prefilter	$a_{lm} \to e^{-l(l+1)\sigma^2/2} a_{lm}$	yes	direction per degree	$O($
T6	band truncation	keep $l \le L$, zero above	yes	every kept coefficient	fr
T7	point inversion	$a_{lm} \to (-1)^l a_{lm}$	yes	P_l	$O($
T8	screw, index shift	rotation $n\gamma$, slide $-2n/N$	no	area	$O($
T9	synthesis	$f(n) = \sum a_{lm} Y_{lm}(n)$	no	none	$O($
T10	gradient	∂_θ, ∂_ϕ by recurrence	$l \pm 1$	none	$O($
T11	translation	degree l draws on degrees 0 to l	**no**	degree 0	$O($
T12	octant indicator	project onto 8 indicators	**no**	total weight	ra
T13	linear interpolation	not an operator on a at all	n/a	vertex values	$O($

T5 and T10 are not implemented. T13 is what the viewer draws.

Three rows carry the weight of the table.

T5 is one addition away. Its action is T4's action with a different time.

T11 is the exception proving the rule. Translation is the only physical move in the table that is triangular in degree instead of diagonal. It is avoided by giving each source its own depth factor in T3 instead of moving an origin.

T13 is in the table to be seen for what it is. It is not a transform of the reading. It is a straight line drawn between two samples of the reading, and the grid measurement below says how far that line sits from the curve it replaces.

3.3 THE COMPOSITION LAW

Rows T3 to T7 are all diagonal. Diagonal operators on the same eigenspaces commute, and composing them multiplies one number per degree. The chain from an interior state to a screen pixel collapses to a single vector of $L + 1$ gains:

$$a_{lm}(\text{screen}) = g_l\, a_{lm}(\text{state}), \qquad g_l = (r/R)^l e^{-l(l+1)\tau_{\text{total}}}, \qquad \tau_{\text{total}} = \tau_{\text{conduction}} + \tau_{\text{pixel}}$$

The two smoothing times add. The heat kernel on the sphere is a semigroup: smoothing for τ_1 then τ_2 is smoothing once for the sum. Nothing about that is an approximation.

The pixel width σ varies per pixel (below, under the footprint), and $\tau_{\text{pixel}} = \sigma^2/2$ enters g, and the gains are per pixel. Group the synthesis by degree, $f = \sum_l g_l S_l$ with $S_l = \sum_m a_{lm} Y_{lm}$, and set $q = e^{-2\tau}$. Since $g_l / g_{l-1} = e^{-2l\tau} = q^l$ up to the depth factor, each degree past the first takes two multiplications, one for q^l and one for g_l. **The filter costs one exponential and** $3L - 2$ **multiplications per pixel**, 28 at $L = 10$.

A correct pixel filter is nearly free. It is one addend inside a τ already being applied. It needs neither a second pass, extra sampling nor a new operator.

3.4 WHY THEY ARE DIAGONAL, AND WHY TWO ARE NOT

The table is not a coincidence and does not have to be memorized. The Laplace-Beltrami operator on the sphere has the degree l subspace as its eigenspace with eigenvalue $-l(l+1)$, that is, $\Delta Y_{lm} = -l(l+1)Y_{lm}$.

Any operator commuting with Δ preserves its eigenspaces and therefore acts on each degree separately.

T3, depth. The field between the source and the boundary satisfies $\Delta f = 0$ in the shell. Harmonic continuation outward is the solution of Laplace's equation there, and separating variables gives $(r/R)^l$ as the radial factor. The exponent is the degree because the degree is the eigenvalue's index.

T4 and T5, smoothing. Both are $e^{\tau\Delta}$, the heat semigroup. An operator built as a function of Δ commutes with it by construction.

T1, T2 and T7, rotations and inversion. These are isometries of the sphere, and Δ is built from the metric. They therefore commute with it. Each degree l subspace is an irreducible unitary representation of SO(3) of dimension $2l+1$, and rotation therefore mixes orders inside a degree while moving no power between degrees. T7's $(-1)^l$ is point inversion, $n \to -n$; a mirror through the equatorial plane gives $(-1)^{l+m}$.

T8 is not an isometry of the sphere. The index shift n turns by $n\gamma$ about the axis and slides the height by $-2n/N$. That is rigid in height and longitude and keeps area, by Archimedes' hat-box theorem [23, *On the Sphere and Cylinder* I, Propositions 33, 42 and 43], but the axial radius $\sqrt{1-y^2}$ changes under the slide. The slide moves power between degrees: $Y_{10} \propto y$ goes to $y - 2n/N$, which has a degree 0 part. Only T8's rotation is diagonal, and P_l is not invariant under T8.

T6, truncation. A spectral projection onto a set of eigenspaces.

T11 does not commute with Δ, and cannot. A translation is not a map of the sphere to itself. There is no Δ on the sphere for it to commute with. Moving a source changes its moments triangularly: degree l of the shifted expansion draws on degrees 0 to l of the old one, and degree 0 stays fixed, as Greengard and Rokhlin's Lemma 2.3 [19, Section 2] gives in the plane. The $O(L^3)$ cost holds by rotating, shifting along the axis and rotating back (Gumerov and Duraiswami [20, Section 1], crediting White and Head-Gordon [50]). Every cheap operation here is diagonal, and the operation avoided is not.

3.5 READING A PICTURE OFF THE COEFFICIENTS

3.5.1 *What a pixel asks for*

A pixel does not ask for the field at a point. It asks for the field averaged over the solid angle it covers [38], and every antialiasing scheme is an attempt to answer that average.

Supersampling estimates the average. Take k samples inside the pixel and average them: cost scales with k, and the error falls about as $1/k$. It converges toward the answer and arrives at it only in the limit.

A band-limited field has the average in closed form. Convolution on the sphere is multiplication in degree. Averaging the field against a zonal kernel K is therefore $a_{lm} \to K_l a_{lm}$, with no sampling anywhere. Choosing K to be the heat kernel of width σ gives T5, and one evaluation of

the prefiltered field is a Gaussian-weighted average over the pixel's footprint, exactly. The exact average over a cap of angular radius θ_0 is also diagonal [14, Part 1], by Hecke's eq. (8) [24, Section 3]: with $x = \cos\theta_0$, $K_l = (P_{l-1}(x) - P_{l+1}(x))/((2l+1)(1-x))$, and $K_0 = 1$ with $P_{-1} = 1$. A square footprint is not zonal and is not diagonal.

The comparison is therefore not between one scheme and a cheaper scheme. Supersampling approximates a quantity this field hands over in closed form.

3.5.2 *The footprint and the time*

The width to filter at is the pixel's own angular size on the boundary, available per fragment from the screen-space derivatives of the direction vector: σ is the length of the change in direction across one pixel, in radians, and $\tau_{\text{pixel}} = \sigma^2/2$.

Two properties follow that a mip chain [51] has to be built to imitate. The width varies per pixel, and the filter is therefore continuous in distance and in surface angle with no levels and no transitions between them. And the width enters τ_{total}, where it is applied by the per-pixel multiplications above.

3.5.3 *Where the degrees stop mattering*

The prefilter also says which degrees can be skipped. $e^{-l(l+1)\tau}$ falls below a threshold ϵ once $l(l+1) > \ln(1/\epsilon)/\tau$, which gives $l_{\max} \approx \sqrt{2\ln(1/\epsilon)}/\sigma$. At $\epsilon = 1/256$, one level of an eight-bit channel, $l_{\max}$ is about $3.33/\sigma$.

This saving is small. $l_{\max}$ reaches down to 10 only when σ is about 0.33 radians. At the center of a disk of radius ρ pixels the direction changes by $1/\rho$ radians per pixel, and $\sigma = 0.33$ gives $\rho \approx 3$, an object about 6 pixels across. Above that size every one of the 121 terms contributes something visible and none may be dropped. The level-of-detail behavior is real, it is derived instead of authored, and it earns nothing at the sizes the viewer is used at.

3.5.4 *The grid against the coefficients, measured*

The viewer samples 121 coefficients at 2,701 vertices of a 36×72 grid, 5.0 degrees to a cell, and the rasterizer fills each cell with two planes. The drawn picture is compared against the field the coefficients hold.

Error per degree, for unit power in that degree alone, as a fraction of the picture's own contrast:

degree	largest	typical	gradient median	gradient 95th
1	0.0010	0.0004	1.25°	5.85°
4	0.0107	0.0026	3.22°	19.15°
7	0.0277	0.0058	5.34°	31.94°
10	0.0592	0.0108	7.47°	47.15°

The largest value error rises 59 times from degree 1 to degree 10, an exponent of 1.77 in the degree. A linear interpolant of a wave of length λ over a step h is wrong by about $(\pi h/\lambda)^2/2$, and

a degree l harmonic has length $360/l$ degrees, an exponent of 2. With $h = 5^\circ$ that estimate gives 0.00095, 0.0152, 0.0466 and 0.0952 at degrees 1, 4, 7 and 10: within the table's rounding at degree 1 and above the measured value at the other three.

field	largest	typical	gradient median
flat spectrum to degree 10	0.0312	0.0050	5.85°
63 deposits at $r/R = 0.80$	0.0273	0.0061	6.67°
63 deposits at $r/R = 0.60$	0.0174	0.0034	6.05°
63 deposits at $r/R = 0.40$	0.0075	0.0015	2.14°

The depth kernel is doing the work in that last column. A deeper reading carries less high-degree power by T3, and the same grid draws it more accurately. The grid is wrong in proportion to the fine structure the state happens to hold.

Read the gradient column. The surface is shaded, and shading takes its normal from the gradient. A linear interpolant has a constant gradient inside a triangle. The drawn normal is a staircase across a field whose gradient turns smoothly. At degree 10 the median normal is 7.5 degrees off and one sample in twenty is more than 47 degrees off. That is the 5-degree quilt visible on the reconstruction, and it is a far larger error than the one percent in the value column.

A per-fragment evaluation has no interpolation error in either column, because there is no interpolation. T10 supplies the exact gradient in closed form for the same reason T9 supplies the exact value: the derivative of the expansion is another expansion.

3.5.5 *What the grid would cost to fix as a grid*

grid	vertices	largest	typical	gradient median
36 × 72	2,701	0.0300	0.0051	5.83°
72 × 144	10,585	0.0077	0.0013	2.89°
144 × 288	41,905	0.0019	0.0003	1.46°
288 × 576	166,753	0.0005	0.0001	0.73°

Four times the vertices for a quarter of the error, the second-order convergence of a linear interpolant, confirmed at 3.95 against a predicted 4.

3.5.6 *The two working sets*

	grid pass	per fragment
samples produced	2,701	as many as there are pixels
basis	read from a table	built by recurrence
bytes touched per pass	1.25 MB	484
samples at 900 px across	not applicable	636,172

The grid pass is bound by a 1.25 MB table that does not fit in cache. A fragment evaluation reads the 121 coefficients and computes the basis from the direction. Its working set is 484 bytes

at any sample count. **No frame time is claimed here.** These are sizes and counts; throughput needs a measurement that is not made here.

3.5.7 *Where this fails*

Anisotropy. The heat kernel is isotropic and a pixel's footprint near the silhouette is stretched. A single σ over-blurs across the short axis. The exact fix wants a directional kernel, which is not diagonal in degree and therefore leaves the composition law behind. The available position is the trade a trilinear filter makes against an anisotropic one.

Edges that are not band-limited. T_5 is exact for a band-limited field, and the boundary field is band-limited by construction, since a reading is a truncated expansion. The room's walls, the beam stops and the arms have real corners, and prefiltering those rings. The split is therefore: the field gets the exact prefilter, the geometry keeps conventional coverage antialiasing. Applying T_5 to the geometry would be a Gibbs artifact presented as a feature.

3.6 INVARIANTS, AND THE SPLIT THEY GIVE

$P_l = \sum_m a_{lm}^2$ is invariant under every element of SO(3) [26], because the degree l subspace is a unitary irreducible representation. The coefficients here are real, and a real a_{lm} has no phase. Under a rotation by α about the read axis the pair $(a_{l,m}, a_{l,-m})$ turns by $m\alpha$ as a vector in the plane: each $|a_{lm}|$ changes, and $a_{l,m}^2 + a_{l,-m}^2$ is invariant. In the complex basis the same rotation moves $\arg a_{lm}$ by exactly $-m\alpha$.

The reading splits exactly along those two properties. **Power per degree is intrinsic**, a property of the source carrying no information about where the observer stands. **Phase is contextual**, a property of the source and the observer together, moving by a known amount under a known rotation. The split needs no estimator, and deflection and torsion are the two halves.

3.7 THE CONSTANTS, AND WHICH KIND THEY ARE

Every constant in the table is a computed number, produced to any requested length by integer arithmetic and verified at 60 places: π, $\sqrt{2}$, $\sqrt{5}$, the golden angle 2.399963229728653 …, and the harmonic unit 0.282094791773878 ….

The golden angle is $\pi(3 - \sqrt{5})$, the index step of T8. The harmonic unit is $1/(2\sqrt{\pi})$, the degree zero harmonic and the scale every other one is built on.

The verification carries no trusted digit string. SHA-256's 64 round constants are defined as the first 32 bits of the fractional parts of the cube roots of the first 64 primes, and its 8 starting words the same for the square roots of the first 8 primes. Recomputing them from the primes and comparing against FIPS 180-4 [13, Section 4.2.2] checks $72 \times 32 = 2{,}304$ independently published bits. All agree. Every copy of the table in the source, fifteen of them, is graded against the definition, and all fifteen match.

Pi is checked twice more, against a 50-digit prefix and against a second computation from Euler's arctangent identity. Machin's and Euler's identities share no term. Agreement between them is not two copies of one mistake.

The measured value this produces. The double the golden placement runs on sits 4.441×10^{-16} radians from the true golden angle. Over 63 indices the accumulated difference is about 2.8×10^{-14} radians, negligible, and a number instead of an assumption.

3.7.1 *The other kind of constant*

The 2022 CODATA adjustment [37, Table XXXII] lists each fundamental physical constant with a value and an uncertainty. About half read *exact*, because the 2019 redefinition of the SI fixed the speed of light, the Planck constant, the elementary charge, the Boltzmann constant and the Avogadro constant by definition. The rest carry an uncertainty no amount of computing will shrink: the fine-structure constant at 1.6×10^{-10} relative, the Newtonian constant of gravitation at 2.2×10^{-5}, every particle mass at its own figure.

No constant of that kind appears in the reading path: no source file of it contains one. The reading engine carries no experimental uncertainty at all. Its only floors are chosen ones, and there are three.

floor	what sets it	how to move it
degree ceiling	the reading's rank, $(L+1)^2$	raise L, and pay $O(L^2)$
number format	double at 16 digits, shader float at 7	carry more digits
calibrated null	a move that changes nothing, measured	a better null

The null is the only measured floor of the three, and it measures the implementation's own arithmetic and not the world. The power-under-rotation null floor stands at 4.005×10^{-16}.

3.8 NOT SETTLED

T5 and T10 are not implemented. The mathematics above is written and the shader is not. Nothing here is a frame time, and the claim that a fragment evaluation is faster than a table read is an architectural argument from working-set sizes, and not a measurement.

The anisotropic case has no answer beyond accepting the isotropic kernel's over-blur.

The $l_{\max}$ constant of $3.33/\sigma$ follows from a threshold of one level in eight bits. The crossover where dropping degrees begins to pay is not measured against a real frame.

T11's cost is quoted as $O(L^3)$ from the shape of the re-expansion and is not timed here, because the readings avoid it.

Depth and degree do not interact in the choice. Depth multiplies a column by about a constant without reordering degrees. The price and the structure are chosen separately. This is measured, and the mechanism is T3's diagonality.

4

THE OCTANT LEXICON

This chapter records what an eight-letter reading of a lit set does and, at greater length, what it cannot do, with SHA-256 compressing one block as the instance.

Eight regions, one letter each, and a word that a computation defines as it runs. The alphabet is general. It reads a lit set on a boundary and knows nothing about what lit it. The same eight letters serve any object that can be placed on a sphere.

The central limit comes first: the alphabet carries eight numbers about a source with 256 degrees of freedom, and the result that looks like a success is close to free. Neither of those is a defect in the method. They are the size of the instrument, counted.

4.1 THE ALPHABET

Choose some regions. A state writes one letter per region, the letter being the share of the lit set sitting inside it. A step of the computation rewrites every letter at once, and the numbers tracked across the steps are the word it defines.

The regions are free. They may sit at any origin, inside the object or outside it, they may overlap each other and anything else, and they may take any shape. On a placement a region reads through its membership vector, the set of points it holds: eight regions holding the same points read rank 1, every letter equal. Geometry enters the reading exactly through membership, and a region's shape counts as far as it decides which points fall inside.

Two properties follow from choices, and they are kept apart from the reading itself. Regions tiling without gap or overlap force the letters to add to one, which costs one degree of freedom. Regions that overlap carry no such constraint and can therefore hold one number each.

4.2 THE INSTANCE THIS CHAPTER USES

Eight regions cut by the sign of each coordinate. Every one has a trilateral right-angle corner at the origin, all eight corners meet at that single point, and they tile space with no gap and no overlap. On the boundary the same split cuts eight congruent spherical triangles, each with three right angles and each of area $\pi/2$ [15], and the measured shares sum to 1.000000000000000.

This is the convenient case and not the general one. It is easy to state, easy to draw and cheap to compute, and every number below inherits the eight and the tiling from that choice.

Nothing above mentions a hash. What lit the set is the caller's business.

4.3 THE SIZE OF THE INSTRUMENT

A reading to degree L carries exactly $(L+1)^2$ real numbers about its source, whatever the source is. Set that against the number of degrees of freedom in the source and the shortfall is the blindness, before any measurement is taken.

degree	coefficients	rank	blind	least singular value
4	25	25	231	4.495
8	81	81	175	4.365
12	169	169	87	3.818
15	256	256	0	3.474×10^{-3}
16	289	256	0	1.524×10^{-1}

Against a 256-bit lit set, a reading to degree eight recovers 81 directions and is blind in 175. The map reaches the rank its coefficient count allows at every degree below the source count. The shortfall in coefficients accounts for the blindness on its own. Neither sample size, precision nor the number of beams changes it.

The eight-letter alphabet here is rank 8, seven free numbers once the weight is fixed, blind in 248 of 256 directions.

Degree fifteen is the floor, where $(L+1)^2$ first reaches 256. Reaching the floor is not reaching a usable reading: the least singular value collapses to 3.5×10^{-3} there and recovers to 1.5×10^{-1} at sixteen. One degree of headroom is worth 44 times in conditioning. A reading degree picked by counting coefficients alone lands on fifteen and reports success.

4.4 THE OBJECT BEING READ

Everything from here down is one instance: SHA-256 [13] compressing a single block, with its 256 working state bits taken as the lit set after each round. The instrument is tried on the hash, and the trying is not the subject.

A frame is the eight working words as the round leaves them, with nothing added back. That distinction decides every number below.

The measurement counts which object it has instead of trusting the caller. A round only copies six of its eight words, b, c, d taking a, b, c and f, g, h taking e, f, g. On the working state all 378 carried words match the words they come from, and on the digest none do.

4.5 THE PLACEMENT UNDER IT

The 256 state bits go onto a golden placement [45, 43]. One index of shift on that placement is a rigid screw: turn 2.399963 radians, axial slide -0.007812, pitch -0.003255258. The pitch holds for every amount. One helix carries them all, and where a shifted bit lands is fixed once the amount is known. The shift is rigid in height and longitude, the coordinates the pitch is read in, and it keeps area by Archimedes' hat-box theorem; it is not an isometry of the sphere, because the axial radius $\sqrt{1 - y^2}$ changes under the slide.

4.6 THE WORD

Compressing the empty message, read to degree eight. The delta [30] is the share of the reading that moved since the round before.

round	oct0	oct1	oct2	oct3	oct4	oct5	oct6	oct7	delta
1	0.142	0.126	0.126	0.102	0.102	0.150	0.157	0.094	0.0%
2	0.122	0.174	0.122	0.078	0.130	0.113	0.113	0.148	12.9%
3	0.171	0.111	0.120	0.137	0.103	0.103	0.145	0.111	14.0%
4	0.111	0.103	0.128	0.120	0.154	0.120	0.103	0.162	12.8%
8	0.156	0.117	0.141	0.133	0.117	0.078	0.102	0.156	8.5%
16	0.135	0.143	0.151	0.151	0.095	0.079	0.111	0.135	10.6%
24	0.137	0.137	0.085	0.111	0.120	0.128	0.137	0.145	7.5%
32	0.095	0.127	0.143	0.095	0.127	0.143	0.151	0.119	10.1%
48	0.136	0.127	0.127	0.085	0.127	0.102	0.153	0.144	12.4%
64	0.150	0.129	0.114	0.121	0.129	0.114	0.143	0.100	5.9%

The digest at the end is `e3b0c442...7852b855`, the published value for the empty message [13]. That check confirms the trace is the hash and not something shaped like one.

4.7 THE DELTA AGAINST ITS FLOOR

The delta averages 11.9% over the first eight rounds and 7.1% over the last eight.

Two unrelated states of the same weight differ by 9.4% in this same alphabet. That figure is the redraw level, the mean delta over 200 pairs of independent states drawn at the matched weight. Without it the delta is a number with no scale.

quantity	value
redraw level	9.4%
delta, first eight rounds	11.9%, or 1.27 times the floor
delta, last eight rounds	7.1%, or 0.76 times the floor
first round under the floor	5
weight range across the run	108 to 140 lit bits

The early rounds move the reading further than two unrelated states do. A reading whose delta clears its own independence level is not reporting counting noise, because counting noise is the level it cleared.

Two cautions bound that.

The fall is a trend in the means and not a monotone series. The per-round deltas over the first eight rounds run 12.9, 14.0, 12.8, 8.4, 11.9, 14.9, 8.5 and 12.0. Round 5 is the first crossing and not the last time the delta sits high.

Naming the move is not part of the measurement. A move carrying mass from octant to octant instead of mixing it will raise the delta above independence, and a rigid relabeling of the indices is such a move. Reading the early excess as the register shift is a reading of why, and it is not

measured. The measurement says only that the excess is present on the working state and absent on the digest.

4.8 HOW MUCH OF EACH CHANGE THE ALPHABET COULD SEE

The delta says the reading moved. It does not say how much of the change the reading was able to move for, and those are different questions. A reading is a linear map, two states give the same letters exactly when their difference lies in its kernel, and the part of a change lying there is invisible however large it is.

The visible fraction answers the second question: $\|Pd\|^2 / \|d\|^2$, the squared share of a difference d lying in the row space [16] of the reading, which for eight indicator arms holds the per-octant totals of the change and discards the rest.

	visible fraction	against an unstructured draw
first eight rounds	0.0397	1.48 times
last eight rounds	0.0138	0.51 times
mean over all 63 pairs	0.0278	1.04 times

Read the split and never the mean. The early changes sit in the directions the alphabet can see above what chance would put there, the late changes sit in its blind directions below what chance would put there, and the mean is those two canceling. Taken alone the mean reports that the reading catches an arbitrary few percent, and that reading of it is an artifact of averaging.

The early to late ratio is 2.88, against 1.68 for the delta over the same split, on 8 pairs at each end. The direction is clear and the sample is small, and the ratio should be quoted with the 8 in the same breath.

The two routes share no denominator. The delta divides counts by a weight moving 4.3 bits a round, and that moving denominator is a possible reason the delta's trend might be bookkeeping instead of arrangement. The visible fraction is a ratio of two norms of one difference vector in raw counts and carries no such term. It shows the trend larger. The denominator damps the effect and does not produce it.

The control arm licenses reading any of this. Differences with no structure in them must return the fraction the row space predicts, and for differences holding the weight that figure is 7 of 255 and not 8 of 256: such a difference sums to zero, the all-ones direction lies inside the row space since the eight indicators add to it, and the difference lives in the hyperplane orthogonal to all-ones. Measured over 200 draws the control returns 0.0267 against a predicted 0.0275. The measurement is withheld whenever the control misses.

4.9 READING THE DIGEST INSTEAD OF THE WORKING STATE

The digest is the wrong object, and reading it gives a different and wrong account of the alphabet. A digest adds the initial values back into the working state, and that feed-forward is a mixing step applied after the round, hiding the round's own structure. Read on the digest sequence, the delta sits at the redraw level from round one, leaving no room to show mixing take hold, and the

arrangement appears to move at a constant rate from the first round to the last. Eight letters then look too coarse for the question.

On the working state the delta does neither: it falls, and it crosses the floor on the way down. The saturation is a property of the digest sequence and not of the alphabet. The count of carried words separates the two objects, all 378 surviving on the working state and none on the digest, and the measurement asserts that count before it reports anything. The limit that holds whatever object is read is the counting statement in the size of the instrument.

4.10 THE DISTINCTNESS RESULT IS CLOSE TO FREE

Sixty-four rounds write sixty-four distinct signatures, a coherence of 100%, with no collisions. This is not evidence that the letters carry meaning.

A signature is eight counts over one weight, and distinctness follows from the counts. At half weight each octant's count of about 32 points is hypergeometric with variance about 7.0, $32 \cdot \frac{1}{4} \cdot 224/255$. Two independent counts agree with probability about 0.106, seven free letters agree with probability about 1.5×10^{-7}, and the 2,016 pairs among 64 signatures expect about 3×10^{-4} collisions [12]. Seven free numbers separate sixty-four arbitrary states nearly always, whether the seven carry anything about the source, and any count of distinct signatures quoted without the rank beside it is quoting that arithmetic.

It is computed because the complement would have been informative. A collision is a null permutation, a move this reading cannot see. None occurs, which rules that out and establishes nothing beyond it.

What would be evidence: a signature that predicts the round it came from.

4.11 THE TWIST

Torsion is the reading that does carry position, and it does not bin anything. A phase recovers a true rotation to 1.8×10^{-13} radians while a magnitude reads a rotation as nothing at all.

The turn recovery assumes the second state is a rotation of the first. Rounds 1 and 64 are not, and the value it returns between them, 4.715210 radians, depends on which coefficient is read first. That twist is a reading at most. It also sits under the same rank bound as everything else here: a phase recovers the turn it is given and does not add coefficients to the reading.

4.12 WHAT A FINER ALPHABET WOULD NEED

Refining the cells raises the coefficient count, and the counting table above is the bar it has to clear. A subdivision leaving more cells than there are points to write into them has bought dead symbols, and one reaching the rank floor has bought a reading whose conditioning collapses.

The trace is checked against the published digest, the eight letters to a sum of one, and the delta to clearing the floor early and falling across the run, and the measurement counts the carried words to confirm which object it reads.

5

A FINER ALPHABET, AND WHAT IT WOULD BUY

This chapter works out what subdividing the octant alphabet would buy before the compute is spent on it, tests a scaling argument for the subdivision, and states what the test leaves standing.

The eight-letter octant reading carries eight numbers about a source with 256 degrees of freedom. The obvious repair is a finer alphabet, and the candidate is to subdivide each octant's spherical triangle [2, 18], giving 8×4^k cells at depth k.

5.1 THE ANSWER

Subdividing does not repair the reading, and the argument settling it is counting, and not statistics. The placement holds 256 points, and past a certain depth there are more cells than there are points to write into them [12].

k	cells	mean placement points per cell
0	8	32
1	32	8
2	128	2
3	512	0.5

The placement points per cell are means. Midpoint subdivision gives unequal cells: at $k = 1$ an octant's center triangle has area 0.551 and each corner triangle 0.340, a ratio of 1.62, and the four sum to the octant's $\pi/2$.

At $k = 3$ there are twice as many cells as placement points, and at least half the cells can never be written at all: the alphabet carries dead symbols. At $k = 2$ the cell count equals the typical lit count, each cell holds two placement points on average, and the reading has stopped being a coarse-graining: it is close to a permuted list of which bits are lit. On a 256-point placement $k = 2$ is the last depth meaning anything, and by then the object being read is no longer a distribution over regions.

Raising the placement's point count moves the ceiling but not the problem, since the redraw level below is set by the ratio of cells to lit points and the subdivision changes that ratio.

5.2 THE SAME CEILING, COUNTED ON THE READING INSTEAD OF THE PLACEMENT

The table above counts cells against placement points. The harmonic reading gives the same ceiling from the other side, and the two agree.

A reading to degree L carries exactly $(L+1)^2$ real numbers about its source. Against 256 degrees of freedom, degree eight is rank 81 and blind in 175 directions, degree fifteen is where $(L+1)^2$ first reaches 256, and the least singular value collapses to 3.5×10^{-3} there before recovering to 1.5×10^{-1} at sixteen. Refinement raises the count of numbers a reading carries; it does not raise the count past the source, and it makes the inversion worse on the way through the floor.

A finer alphabet is therefore bounded twice, once by how many cells the placement can fill and once by how many numbers the degree can carry, and neither bound is statistical.

5.3 THE SCALING EXPECTATION

Separately from the ceiling, a finer alphabet is expected to raise the delta and the redraw level together, leaving their ratio where it was.

The floor half of that is solid. Two independent states of the same weight give a level growing as $\sqrt{c/n}$ for c cells and n lit points, ordinary redraw statistics needing no assumption about the function being read, while each cell holds many lit points. As the points per cell fall toward one, the level saturates at the set difference share, about 0.5 (the units note below), and stops growing. The delta half is only an expectation: a round's movement acts like a partial redraw and is expected to scale the same way.

5.4 A SCALING ARGUMENT THAT DOES NOT HOLD PER PAIR

The argument is that the delta equals the floor times $\sqrt{s}$, where s is the fraction of the lit set a round sends to fresh positions, on the grounds that points which did not move cancel cell by cell and leave the same calculation over sn points. The cell count then cancels between the two expressions and the ratio depends on s alone.

The gap. A share is a count divided by the weight, and the weight changes every round. On the working state it ranges from 108 to 140 lit bits and moves by 4.3 bits per round on average, 3.5% of the weight. A change in the denominator moves all eight shares even for points that never moved. The cancellation the argument rests on does not happen.

The test. Predicting s from the ratio squared and counting the lit points that actually moved, over all 63 consecutive pairs, on the working state and on the finalized digest:

quantity	working state	finalized digest
correlation of predicted against counted	0.105	0.068
worst predicted s	3.633, an impossible value	3.508
observed ratio, per pair	0.382 to 1.906	0.507 to 1.873

As a per-pair estimator it fails on either object, and it returns values above one where no fraction can go. Changing the object moves every figure and changes no conclusion.

What holds. On the mean, with both sides in the same units, the argument's central claim stays close. The counted relocation in the argument's own units averages 1.0172, whose square root is 1.0085, against an observed mean ratio of 1.0361. The two agree to 2.7%.

The units matter: a set difference between two states of weight n_2 out of N saturates at $1 - n_2/N$, about one half, and never at one, and a raw set difference of 0.5 already means complete independence. Comparing $\sqrt{0.5} = 0.707$ against a ratio near 1.0 compares two different quantities.

The mean prediction survives and the per-round estimator does not. Neither outcome supports using this argument as the reason a finer alphabet fails, and the two ceilings above are the reasons to give. Reading s off aggregate percentages instead of per pair gives values that bracket the true one by accident, and those values carry no information about s.

5.5 THE RELOCATION IS CLOSE TO CONSTANT

quantity	working state	finalized digest
set difference between consecutive states	mean 0.5215, range 0.421 to 0.619	mean 0.4969
the same in matched units, where 1.0 is independence	1.0172	0.9976
bits differing between consecutive states	mean 129.9 of 256	mean 127.7
what a coin would give	128.0	128.0

Every round relocates the lit set about as completely as an independent draw would, from the first round onward, with the working state sitting 1.7% above independence and the digest 0.2% below it. Reading s off the ratio is therefore estimating something that barely varies, and the backwards reading carries correspondingly little.

5.6 WHAT A FRAME IS

A frame is the eight working words as the round leaves them, with nothing added back. The finalized digest adds the initial values back into the working state, and that feed-forward is a mixing step applied after the round.

The two objects are told apart by counting and not by assertion. A round only copies six of its eight words [13], b, c, d taking a, b, c and f, g, h taking e, f, g. Over sixty-three transitions that is 378 carried words, and on the working state all 378 match the words they come from while on the digest none do.

The register shift is therefore present in the working state exactly, since being a register shift consists of matching all 378 carried words. A test that undoes the word shift on digest frames finds 108.9 of 224 bits differing against a coin's 112.0, with no exact match in sixty-four rounds, and reports the shift as invisible; that test measures the feed-forward.

What is not established is a link between the shift and the octant delta. The delta's early excess over the redraw floor is measured, and a rigid relabeling of indices is the kind of move producing

such an excess, but attributing the excess to the register shift is a reading of why and is not measured. The two statements sit at different levels and only the first is a count.

5.7 WHAT WOULD SETTLE THE SUBDIVISION

One run at $k = 1$ and $k = 2$, each with its own redraw level computed at the matched weight, reporting the ratio and not the delta. Two ratios near the $k = 0$ value confirm the expectation. A ratio falling with k refutes it cheaply, and the two ceilings still stand either way.

6

THE SHIFT REGISTER, READ AS A DEFORMING SURFACE

A harmonic reading of the SHA-256 compression state renders a surface, and that surface deforms as the round clock turns. This chapter states what the deformation is, shows that it belongs to the state and not to the observer or the probe, and traces it to the word shift of the compression function's step. The observer-invariance making the reading trustworthy is stated first, because it is the calibration letting the rest be read as measurement and not as viewpoint.

6.1 THE OBJECT

SHA-256 compresses in sixty four rounds over a state of eight thirty-two bit words, which is 256 bits. The reading places those 256 bits on 256 points of a ring arrangement and expands the lit set in spherical harmonics. The state becomes a field on a sphere, and the field has a shape. Bit i lights point i. The placement is ring-major and the lit set is word-major, and together they put word s on ring s; the mechanism and the scope below depend on that choice.

The compression agrees with the published digest of the empty string [13] before any of this is read. The object under the instrument is SHA-256 and not an approximation of it.

6.2 THE CALIBRATION: WHICH PART OF THE SHAPE IS THE OBSERVER'S

A shape read this way has two anisotropies, and they answer to a rotation of the observer differently. This is standard representation theory of the rotation group, stated here because it is the null the measurement rests on, and not because it is new.

Deflection, the power per degree $P_l = \sum_m |a_{lm}|^2$, is invariant under every rotation in SO(3): the degree-l subspace carries a unitary irreducible representation of the group [11], and a magnitude built from it cannot record how the object was turned [26]. Measured null under a whole-step rotation: $\mathbf{2.539 \times 10^{-14}}$, the arithmetic and not the object.

Torsion, the phase $\arg a_{lm}$ of the same coefficients, moves by exactly $-m\alpha$ under a rotation by α about the polar axis. It records the turn, sign included. A rotation about any other axis mixes the orders m within each degree, and one coefficient's phase no longer records the turn alone. Measured recovery of a known angle: $\mathbf{1.803 \times 10^{-13}}$ radians.

The magnitude of the anisotropy is therefore observer-independent and its direction is observer-dependent, and the split is exact. This is a property of the reading and holds for any lit set, SHA-256 or otherwise. Its role here is precise: a change in the shape's **magnitude** across the clock is a change in the object, while a change in its **direction** could be either the object or the

frame. Every claim below is made on a magnitude, where the observer has been divided out, or is checked against a control holding the frame fixed.

6.3 THE MEASUREMENT, ON THE SURFACE ITSELF

The reading renders exactly one deformed surface in the scene, and it is the state's. Every enclosing shell measures a vertex-radius standard deviation of exactly zero, and each is a perfect sphere. The state's surface measures a standard deviation of 0.999 against a mean radius of 8.795, at 11.4 percent. Read as a radius field over the sphere and tracked across the sixty four rounds:

feature	degree	across the clock
the elongation, a lemon	2	$q_{zz} \approx -1.50$ at every round, 17 percent of the mean radius
the offset, a cone	1	d_z swings about 0.07 and changes sign several times

The quadrupole is a fixed feature: the surface is elongated by the same amount, in the same axis, the whole way through. The dipole migrates and inverts. The cone walks around the surface and changes which pole it favors more than once.

6.4 THE CLAIM, STATED AS A NUMBER

The dipole does not merely move. It moves **coherently**: its sign, taken round by round, comes in runs, and not in the alternation independent signs would give. Twenty runs against an expected 30.5, at a run-length test [46] assuming nothing about the values themselves:

$$z = -2.87$$

Measured a second way, on a different object through different code, the rendered mesh in the viewer against the lit set, the same statistic reads $\mathbf{z} = -\mathbf{3.0}$ to one decimal. Two pipelines with no shared arithmetic agree within two tenths.

A surface computed from a slowly changing state drifts at that level. The next section is the control pinning the change to the state.

6.5 PINNING THE CHANGE TO THE STATE

Two interventions, because the round clock advances everything at once and an observation that the picture changes when the clock moves does not by itself say what moved it.

Moving the probe changes nothing. Holding the round fixed and swinging the beam 180 degrees in azimuth, 128 in elevation and 4.7 times in radius leaves the surface unchanged in every digit: mean radius, dipole and quadrupole all identical before, during and after. One round of the clock, with the beam untouched, moves the dipole by five times. The surface answers to the state and not to the probe.

Replacing the rounds with independent states removes the effect. The real round sequence gives $z = -2.87$. A control swapping the rounds for independent states of the same per-round weights, everything else identical, gives $z = 0.00$ with a spread of 0.94. The real sequence sits 3.1 control deviations out. The drift is in the succession of states, and not in the reading of any one of them.

6.6 THE MECHANISM, WHICH IS NOT EXOTIC

SHA-256's round computes two words and shifts the other six [13, Section 6.2.2]:

$a' = T_1 + T_2$	$e' = d + T_1$
$b' = a$	$f' = e$
$c' = b$	$g' = f$
$d' = c$	$h' = g$

Six of the eight words at round r are round $r-1$'s words in a new position, measured at 63 of 63 rounds. A surface computed from a state that is three quarters carried over cannot jump between consecutive rounds, and a coherent, autocorrelated drift is therefore the shift register showing.

The run count follows from the shift with a number. The dipole is zonal, and word s sits on ring s at colatitude $\pi(s + 1/2)/8$. A fresh word enters at a or at e and rides four rings over four rounds. With A_r and E_r the popcounts of the two fresh words at round r, and $c_j = \cos\theta_j = 0.981, 0.831, 0.556, 0.195$ for $j = 0, \dots, 3$, the dipole is $d(r) = \sum_j c_j A_{r-j} - \sum_j c_{3-j} E_{r-j}$. Taking the fresh popcounts independent with equal variance, its correlations at lags 1, 2 and 3 are 0.693, 0.354 and 0.096, and zero beyond. By Sheppard's rule [40] a sign change between neighbors has probability $\arccos(\rho_1)/\pi = 0.256$, about 17 runs in 64, against 20 measured and 30.5 for independent signs. The $z = -2.87$ reproduces from 41 and 23 signs of 64: the expected run count is 30.47 and its standard deviation 3.65 [46].

The reading is built to show a field, and the word shift of SHA-256's step function appears in it: legible to the eye as a lemon whose cone walks and inverts, and holding at three standard deviations when the drift is measured against a control. The result is not a new property of SHA-256. It is a demonstration that a harmonic boundary reading makes a known structural property of the compression function visible and measurable, with the observer divided out and the cause established by intervention. The chaining of compression calls across message blocks [36, 8] is a different structure, and one block's compression holds none of it.

6.7 SCOPE

The observer-invariance is classical. Rotation-invariant power and orientation-carrying phase are standard SO(3) representation theory, the same fact behind an angular power spectrum being usable without its phases. It serves here as the calibration and finds nothing new.

The persistent quadrupole is the placement's. With word s on ring s, the zonal quadrupole is proportional to $\sum_s (3\cos^2\theta_s - 1)\,\mathrm{pop}_s$. The ring coefficients are 1.886, 1.074, −0.074 and −0.886, mirrored in the southern half, and they sum to 4.0, where the sphere's area average of $3\cos^2\theta - 1$ is zero. At a mean popcount of 16 the expected term is 64 units. With a popcount variance of

8 per word the fluctuation is about 9.4 units, and the sign holds at every round with a wobble near 15 percent, for random states as much as for SHA-256's. Scoring the degree-2 power against shuffled lit sets on the same placement separates SHA-256 from the placement, and that score is not reported here.

What is specific to SHA-256 is the dipole inverting in runs at $z = -2.87$, and its trace to the word shift by intervention and by the run count it predicts.

No claim is made about a weakness in SHA-256, about its output, or about anything a cryptographic reduction would touch. The reading is taken on the intermediate compression state, which carries the shift structure by design. The digest is the state after round 64 with the block's chaining input added word by word [13, Section 6.2.2, step 4], a single addition after the last round, and it is a different object and is not read here.

7

ROUND COUNT AS DEPTH

This chapter specifies an experiment measuring how fast the interior stops reaching the boundary as a function of how many rounds it passed through, and where that curve crosses below the detection floor.

This chapter specifies work and does not report it. Everything below is a design. The observables, the compression truncated at an arbitrary round and the driver are not implemented, and no curve is measured. A specification written down before the run is the only kind that can be checked against the run afterwards.

7.1 THE IDEA IN ONE LINE

The boundary readings measure information reaching a surface as a function of depth, and find it falls as $(r/R)^l$. Here depth is the round number. The question is the same one: how much of the input reaches the output, measured against how far in it started.

7.2 WHY THIS IS THE RIGHT QUESTION AND BREAKING IT IS THE WRONG ONE

A break needs a boundary observable departing from flat by more than sample noise, at full rounds. The strict avalanche [48], bit bias and Rényi [39] observables read flat at full rounds. Linear [34] and differential [4] cryptanalysis reach part of the way. Thirty-one steps is the collision record [35], with a colliding pair found in practice [32]. Other attacks reach further: second-order differential collisions on reduced SHA-256 [5], semi-free-start collisions at 39 steps [31], and biclique attacks at 45 steps for preimages and 52 for pseudo-preimages [27]. None reaches 64. Hunting the full-round hole is hunting a thing the observables already read as absent.

Measuring the decay is different. Reduced rounds leak, and the leak is not written down as a curve in the depth variable with a detection floor drawn on the same axes. That curve is a real result. It quantifies the difficulty instead of asserting it, and it makes a falsifiable claim: the leak at round k is this size, and here is where it falls under what this many samples can see.

7.3 WHAT IS MEASURED

Run SHA-256 [13] truncated at k rounds, for k from 1 to 64, and read a boundary observable at each.

Strict avalanche. Flip one input bit, count how often each output bit flips, over many inputs. A flat function sits every counter at half the sample size. The departure from half, summed across output bits, is the leak.

Bit bias. How far each output bit's one-rate departs from half.

Collision entropy of the output byte histogram.

Each returns a single number per round count. The run is one curve per observable, 64 points each.

7.4 THE DETECTION FLOOR, DRAWN ON THE SAME AXES

A bias of size b needs about $1/b^2$ samples to see above the noise. At a flat function one counter's rate over N samples has standard deviation $1/(2\sqrt{N})$. The strict avalanche observable sums |departure| over M counters, and that sum has a null mean of $M\sqrt{2/\pi}/(2\sqrt{N})$, M times the mean of a half-normal. That floor is a horizontal line on the same plot as the decay curve, and where the curve crosses under it is the round count past which this run cannot tell the function from flat.

The floor is not a nuisance to be hidden. It carries the result: it says what was in reach of this much compute, and a later run with more samples can lower the line and show the curve continuing underneath, or show it genuinely flattening.

7.5 THE ARITHMETIC

The decay spans many orders of magnitude, from a large leak at a few rounds to nothing measurable at full rounds. The counts are integers. A 64-bit counter holds them exactly to 1.8×10^{19}, where single precision holds every integer exactly only up to 2^{24}. With integer counters the curve stays a curve down to where sample size and not arithmetic stops it, and measuring the decay is kept separate from measuring a floating-point floor.

7.6 WHAT EACH OUTCOME WOULD MEAN

The curve falls smoothly and crosses the floor around the round count cryptanalysis already reaches. The expected result. It turns the known reduced-round weakness into a measured decay rate and puts a number on the margin.

The curve has a step, a shoulder, or a bump that does not fall. A round range where more structure survives than the smooth decay predicts. That is where a cryptanalyst would look, and the measurement would have pointed at it instead of guessing. Almost certainly it will not happen at full rounds, and the experiment would show it if it did.

The curve is already under the floor by a few rounds. The samples were too few. Lower the line with more samples and run again. Nothing is concluded from a curve starting under its own floor, and a run passes that check before its flat tail means anything.

7.7 WHAT THIS DOES NOT CLAIM

It does not claim a break, a bias at full rounds, or a shortcut to a preimage. It claims a measurement: the leak as a function of depth, with the floor saying how far down the measurement could see. A flat tail under a floor is not evidence the function is flat there. It is evidence the run could not see a leak that small, a statement about the run and not about SHA-256.

7.8 WHAT HAS TO BE BUILT

A compression function that stops at an arbitrary round k, the three observables computed on it, a driver collecting one curve per observable with a round-count parameter, and a plot carrying the floor as a second series.

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